

Dynamical Phase Transitions in Nonequilibrium Networks

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Dynamical phase transitions (DPTs) characterize critical changes in system behavior occurring at finite times, providing a lens to study nonequilibrium phenomena beyond conventional equilibrium physics. While extensively studied in quantum systems, DPTs have remained largely unexplored in classical settings. Recent experiments on complex systems, from social networks to financial markets, have revealed abrupt dynamical changes analogous to quantum DPTs, motivating the search for a theoretical understanding. Here, we present a minimal model for nonequilibrium networks, demonstrating that nonlinear interactions among network edges naturally give rise to DPTs. Specifically, we show that network degree diverges at a finite critical time, following a universal hyperbolic scaling, consistent with empirical observations. Our analytical results predict that key network properties, including degree distributions and clustering coefficients, exhibit critical scaling as criticality approaches. These findings establish a theoretical foundation for understanding emergent nonequilibrium criticality across diverse complex systems.

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Introduction—Dynamical phase transitions (DPTs) are critical phenomena that unveil critical behavior emerging over time and represent a new frontier of nonequilibrium physics [1–10]. Distinct from conventional phase transitions, typically driven by external parameters such as temperature or pressure, DPTs occur at a critical time t_c , where physical observables display nonanalytic behavior. While these transitions have been extensively studied in quantum systems, their role in classical contexts remains largely underexplored, with only limited progress emerging from mean-field analyses of off-equilibrium relaxation toward equilibrium [11,12] and from simulations [13].

Intriguingly, recent experiments on macroscopic complex systems, including ecological collapse [14,15], financial crises [16,17], and power grid failures [18], report abrupt dynamical changes at finite times. These events, though not formally classified as DPTs, share key characteristics such as abrupt, collective changes. The striking similarity between these phenomena and quantum DPTs suggests that critical temporal dynamics may be a widespread feature of complex systems. This raises intriguing questions about their broader relevance and underlying universality.

More recently, studies on social networks have presented compelling evidence for the emergence of DPTs in classical settings [19–22]. These experiments reveal that some physical observables $z(t)$ diverge at a critical time t_c according to a universal hyperbolic scaling,

$$z(t) \sim (t_c - t)^{-1}. \quad (1)$$

Unlike quantum DPTs, where nonanalyticity emerges without divergence, this classical hyperbolic scaling reveals a distinct signature of temporal criticality, appearing to be ubiquitous across distinct domains. For instance, hyperbolic discounting observed in economics [23–26], which rationalizes procrastination and deadlines, also adheres to this scaling, as evidenced by surges in academic submissions near deadlines [27]. Despite its prevalence, the theoretical underpinnings of this universality remain poorly understood.

Existing theoretical frameworks for nonequilibrium classical phase transitions generally fit into two broad categories. The first addresses different phases emerging either under time-varying control parameters akin to equilibrium transitions [21,28–31]. The second encompasses models of self-organized criticality [32–36], where critical points occur in the asymptotic limit $t \rightarrow \infty$ rather than at a finite t_c . Neither category fully captures the finite-time singularities characteristic of DPTs, indicating a need for new theoretical approaches.

In this Letter, we introduce a theoretical framework that predicts the emergence of DPTs in nonequilibrium networks. Our central insight is that nonlinear interactions play a pivotal role. Drawing an analogy to many-body physics, a random graph without interactions behaves like a system of noninteracting particles with no DPTs. Once nonlinear interactions are introduced, however, the system's collective dynamics are fundamentally altered. To demonstrate this, we consider a simple model involving quadratic interactions, motivated by triangle closure, a

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ubiquitous mechanism in social networks. Within this model, quadratic interactions between edges naturally yield DPTs. Remarkably, the model can be solved analytically, reproducing the universal scaling behavior expressed in Eq. (1). Furthermore, we demonstrate that critical network characteristics, such as the degree distribution and clustering coefficient, exhibit critical scaling behavior as t_c approaches. These results establish a theoretical basis for understanding nonequilibrium criticality in classical systems, shedding light on parallels with quantum DPTs.

Theoretical framework—We consider a directed network comprising N vertices undergoing random edge addition and removal. Specifically, a vertex i connects to a vertex j to form a new edge with a probability rate α/N , while edges are removed at a rate γ . The parameters α and γ characterize the rates of edge formation and removal, respectively, and the factor $1/N$ ensures the sparsity of the network. The evolution of the expected average degree $z(t) := (1/N)\langle\sum_{ij} A_{ij}\rangle$, where $\langle\ldots\rangle := \int \ldots P(A, t) dA$, follows the linear differential equation,

$$\frac{dz}{dt} = \alpha - \gamma z,$$

with the stationary solution given by $z(t \rightarrow \infty) = \alpha/\gamma$. In this noninteracting scenario, the network remains a random graph, exhibiting no DPTs.

Next we will show that the emergence of DPT requires nonlinear interactions. A natural approach is to modify the edge addition rate to $\alpha/N + U_{\text{int}}$, where $U_{\text{int}}(A)$ represents an interaction term dependent on the Boolean adjacency matrix A . While the edge removal rate could also be modified similarly, we hold this parameter constant for simplicity. The interaction term can be expanded as $U_{\text{int}}(A) = \beta_0 A + \beta A^2 + O(A^3)$. To isolate the nonlinear effects, we neglect the linear term and retain only the leading nonlinear contribution, yielding $U_{\text{int}}(A) \approx \beta A^2$, where β is the coupling constant. These interactions are naturally interpreted as a triadic closure mechanism [37]: two vertices, i and j , sharing a common neighbor are more likely to form a direct connection. This occurs with an additional rate proportional to $\sum_k A_{ik} A_{kj} = (A^2)_{ij}$, leading to the following network dynamics:

$$W[A_{ij}(0 \rightarrow 1)] = \left(\frac{\alpha}{N} + \beta(A^2)_{ij} \right), \quad (2a)$$

$$W[A_{ij}(1 \rightarrow 0)] = \gamma. \quad (2b)$$

While triadic closure is typically associated with social networks, our theory treats it as the leading nonlinear interaction, extending its applicability beyond social contexts.

In the weak-interaction regime (β small), the network behaves as a random graph with no DPTs. However, as β

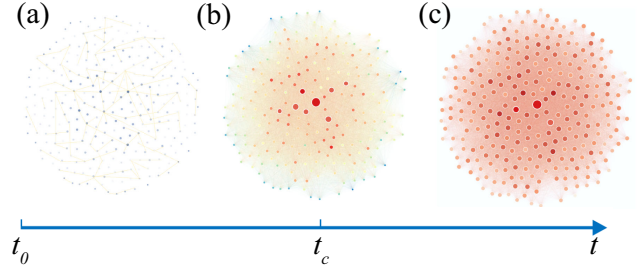


FIG. 1. Illustration of network evolution over time for $\beta = 0.5$, $\alpha = \gamma = 1$ with $N = 300$: (a) Before the critical time t_c , the network exhibits a sparse structure. (b) At $t \rightarrow t_c$, the network transitions to a scale-free topology. (c) After t_c , the network becomes dense, resembling a nearly complete graph.

increases, nonlinear interactions significantly alter the dynamics, leading to edge clustering and finite-time singularities. Figure 1 illustrates the time evolution of the network for the proposed model for $\beta = 0.5$, and $\alpha = \gamma = 1$, which suggests that there is a critical time t_c where the network structure changes significantly.

To quantify this rapid change, we derive the time evolution of the density of arbitrary connected subgraphs, $\rho(G)$, which is given by the Schwinger-Dyson (SD) equations under the sparsity condition (see Supplemental Material [38], Sec. 1.1). For the simplest subgraph, a link between two vertices, the density corresponds to the average degree z , following

$$\frac{dz}{dt} = \alpha - \gamma z + \beta(z^2 - \Delta), \quad (3)$$

where $\Delta := (1/N)\langle\sum_{ijk} A_{ij} A_{jk} A_{ik}\rangle$ measures the triangle density in the network. The kinetic term $\alpha - \gamma z$ is the same as the noninteracting random graph model, whereas the interaction term $\beta(z^2 - \Delta)$ introduces a feedback loop that amplifies clustering.

This differential equation is not closed because Δ depends on the number of four-vertex rectangular shapes. These four-vertex rectangular shapes, in turn, depend on the number of five-vertex shapes, and so forth. This hierarchy creates an infinite tower of equations, a hallmark of the SD equations. For instance,

$$\frac{d\Delta}{dt} = \beta(z^2 - \Delta) - 3\gamma\Delta + O(\beta^3), \quad (4)$$

where $O(\beta^3)$ represents the terms with higher orders, such as four-vertex shapes. Applying a standard perturbative truncation scheme to the SD equations and noting that $\Delta \sim O(\beta)$, we can omit it in Eq. (3), resulting in

$$z(t) \sim \begin{cases} \text{constant} & (\beta \leq \frac{\gamma^2}{4\alpha}, t \rightarrow \infty), \\ 1/(t - t_c) & (\beta > \frac{\gamma^2}{4\alpha}, t \rightarrow t_c), \end{cases} \quad (5)$$

where the critical time t_c is given by $t_c = \omega^{-1}[\text{acot}(2\omega) + (\pi/2)]$ and $\omega = \sqrt{|\alpha\beta - (\gamma^2/4)|}$. Thus, t_c depends on (α, β, γ) only through ω , and a finite-time singularity occurs only in the supercritical regime $\beta > \gamma^2/(4\alpha)$ (see Supplemental Material [38], Sec. 1.1). Moreover, Eq. (5) identifies two distinct phases: In the “equilibrium phase,” when $\beta \leq (\gamma^2/4\alpha)$, $z(t)$ approaches a constant value as $t \rightarrow \infty$, indicating that the network converges to a stationary state. In the “nonequilibrium phase,” when $\beta > (\gamma^2/4\alpha)$, the model predicts a DPT in which $z(t)$ diverges hyperbolically as $t \rightarrow t_c$, consistent with the empirical law in Eq. (1). Importantly, observables other than the average degree diverge with different exponents that reflect their structural order. For example, the triangle density scales as $\Delta \sim z^2 \sim [1/(t_c - t)^2]$, capturing its second-neighbor nature. More generally, higher-order subgraph densities diverge as powers of z , with exponents set by their effective neighborhood order.

The derivation above represents a special case of the infinite tower of the Schwinger-Dyson equation, which takes the adjacency matrix $A(t)$ acting as a field variable in quantum field theory and omits the high-order $\square \sim O(\beta^3)$ (see Supplemental Material [38], Sec. 1.1). Note that adding higher-order terms does not alter the physical picture described above, but introduces corrections to the coupling constant. For instance, considering the contribution of Δ from Eq. (4), while omitting $O(\beta^3)$ terms and applying the adiabatic approximation, results in a correction $\beta \rightarrow \beta' := (3\beta\gamma/3\gamma + \beta) = \beta - \beta^2/(3\gamma) + O(\beta^3)$. This can be interpreted as a dressed coupling constant due to higher-order corrections. Replacing β with the dressed value β' in Eq. (5) provides a precise prediction of $z(t)$, as shown in Fig. 2(a). While the adiabatic approximation is expected to fail for rapid changes, especially as $t \rightarrow t_c$, we find that even this simple approximation agrees remarkably well with numerical simulations for both the equilibrium and nonequilibrium phases. Our theoretical analysis above can be summarized in the phase diagram in terms of α/γ and β/γ , where the phase boundary $\beta'(\beta)/\gamma = (4\alpha/\gamma)^{-1}$ separates the equilibrium and nonequilibrium phases. Figure 3(a) compares our theoretical prediction with numerical simulations, where the dashed line represents the theoretical prediction at the tree level ($\beta' = \beta$), and the solid line incorporates the triangle correction ($\beta' = (3\beta\gamma/3\gamma + \beta)$). We find that both predictions agree well with the numerical results for small β values, whereas the triangle correction agrees with the numerical results even for relatively large β .

Surprisingly, our theory indicates that even small β values can lead to the nonequilibrium phase, provided the kinetic rate α is sufficiently large. This implies that introducing arbitrarily small nonlinear interactions to the random graph model can fundamentally alter the nature of the underlying network evolution, potentially leading to DPT.

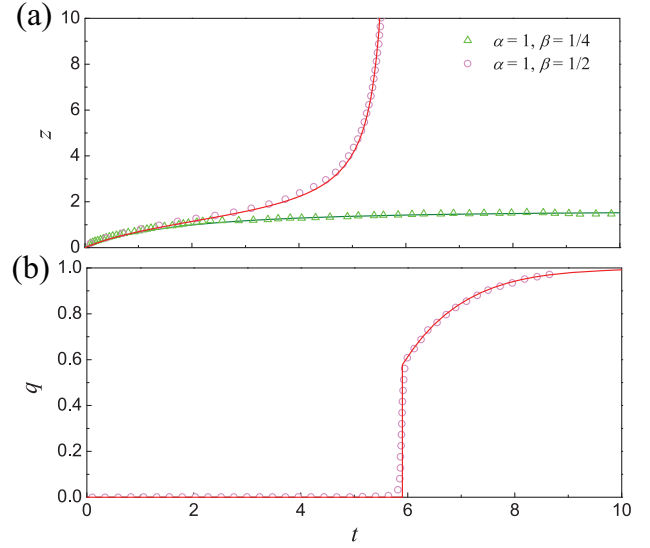


FIG. 2. Theoretical predictions versus numerical simulations for (a) the average degree $z(t)$ and (b) the order parameter $q(t)$, with parameters $\alpha = 1$, $\gamma = 1$, $\beta = 1/4$ (green), and $\beta = 1/2$ (red). Solid curves represent theoretical predictions, while scatter points correspond to simulations for $N = 2000$. For $\beta = 1/4$ (green), the average degree $z(t)$ converges, while the order parameter $q(t)$ vanishes, indicating an equilibrium phase. For $\beta = 1/2$ (red), $z(t)$ exhibits hyperbolic scaling and diverges at a finite critical time $t_c \approx 5.9$. Correspondingly, $q(t)$ shows a sharp jump at t_c , signifying a first-order DPT from a sparse to a dense network.

First-order phase transition—To further characterize the nonequilibrium phase, we define the order parameter $q = \lim_{N \rightarrow \infty} z/N$, representing the fraction of edges relative to a complete graph. By construction, $0 \leq q \leq 1$. In the sparse graph regime [$z \sim O(1)$], we have $q = 0$, which holds both in the equilibrium phase and in the nonequilibrium phase before the critical time $t < t_c$. However, for

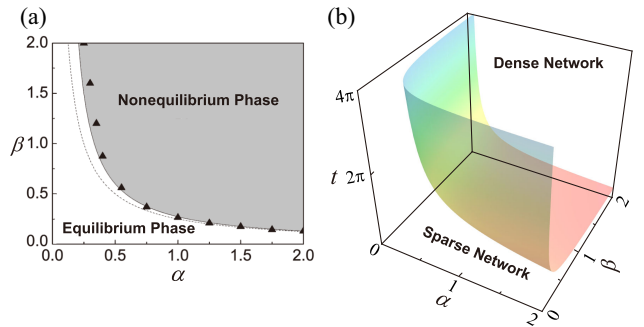


FIG. 3. (a) Phase diagram of the model in the α - β parameter space, showing two distinct phases: equilibrium and nonequilibrium (gray region). The curves represent theoretical predictions, while scatter points correspond to simulation results for the phase boundary. (b) Phase diagram illustrating the transition from sparse to dense graph phases. The colored surface quantifies the phase boundary. Here, α , β , and t are rescaled by γ to render them dimensionless.

$t > t_c$, the network becomes dense, resulting in $q > 0$. Such a sparse-dense network phase transition has been observed in theoretical models and real-world systems, including network densification [39] and clique percolation [40]. These systems share clustering mechanisms akin to triangle closure, where sparsity collapses when clustering strength exceeds a critical threshold. However, conventional models display phase transitions controlled by external parameters. In contrast, in our model, this transition is dictated by a finite critical time, marking it as a DPT. The singularity at $t = t_c$ in Eq. (5) indicates a first-order phase transition, characterized by a discontinuity, i.e., a latent heat, in $q(t = t_c)$ from zero to a nonzero value. This behavior is illustrated in Fig. 2(b).

The analytical result in Eq. (4) is valid only for $(t < t_c)$, as the sparsity condition breaks down beyond the critical time. To determine $q(t)$ for $t > t_c$, we need a better understanding of the observed DPT. For sufficiently large coupling, the network starts as sparse, with kinetic terms dominating the dynamics. As connections grow, nonlinear interactions driven by triadic closure become increasingly significant, creating competition between kinetic terms and nonlinear interactions. Once nonlinear interactions dominate, a rapid cascade of edge formation occurs, resulting in a DPT.

After the DPT, the network condenses into a tightly connected core forming a complete graph, accompanied by a few isolated vertices with degree zero. Assuming the fraction of isolated vertices is p , the fraction of the complete-graph core is $[1 - p(t)]N$, and thus $q(t) = [1 - p(t)]^2$. As soon as an isolated vertex establishes a single edge to the core, it immediately connects to all other vertices in the core because the number of common neighbors is $O(N)$, which diverges. Consequently, the rate at which an isolated vertex connects to a vertex in the core increases. This implies that $p(t) = p_0 e^{-\alpha(t-t_c)}$ decays exponentially due to the random connections to the core, where p_0 is the fraction of isolated vertices at $t = t_c$. From here, we obtain

$$q(t) = \left(1 - p_0 e^{-\alpha(t-t_c)}\right)^2, \quad (6)$$

for $t \geq t_c$. This prediction agrees well with the numerical simulations [Fig. 2(b)]. The latent heat $q(t_c) = (1 - p_0)^2$ is related to the critical fraction of isolated vertices p_0 , as we will discuss shortly.

In summary, for the equilibrium phase, the network will remain sparse for any t . For the nonequilibrium phase, the critical time t_c further separates the states into sparse ($t < t_c$) and dense networks ($t > t_c$). The critical time t_c depends on α , β , and γ , where relatively larger β corresponds to a smaller t_c , i.e., the system reaches the critical time sooner. Figure 3(b) illustrates these findings in the three-dimensional phase diagram. The colored phase

boundary surface separates the domains corresponding to sparse and dense networks.

Critical behavior—To investigate the critical behavior near the DPT, we are particularly interested in the degree distribution. For the noninteractive model ($\beta = 0$), the degree follows a Poisson distribution. For finite nonlinear coupling, at the tree level, we find the time-dependent degree distribution $P(k, t)$ satisfies

$$\begin{aligned} \frac{\partial}{\partial t} P(k, t) = & [\alpha + \beta(k-1)z(t)]P(k-1, t) \\ & - [\alpha + \beta k z(t) + \gamma k]P(k, t) \\ & + \gamma(k+1)P(k+1, t). \end{aligned} \quad (7)$$

This equation can be solved analytically using the generating function method (see Supplemental Material [38], Sec. 1.2). For the nonequilibrium phase, the degree distribution becomes a power-law form as the system approaches the critical time t_c , as shown in Figs. 4(a) and 4(b) for both theoretical predictions and numerical simulations. Eventually, $P(k, t \rightarrow t_c) \sim k^{-\gamma}$ forms a scale-free network [41,42], with the exponent $\gamma = 1$, suggesting the divergence of the average degree z , a signal for a DPT. This is particularly interesting because we do not use mechanisms like preferential attachment to generate the scale-freeness [41,42]. Instead, the critical behavior near the DPT automatically generates scale-freeness, which lies beyond the conventional paradigm of scale-free networks.

Moreover, while the tree-level approximation only applies to $t \leq t_c$, this analytical result allows us to compute the fraction of isolated vertices $p_0 = P(0, t_c)$ (see Supplemental Material [38], Sec. 1.2). Figure 4(c) shows the proportion of isolated vertices p_0 under different coupling constant β while keeping $\alpha = \gamma = 1$ fixed. The results indicate that stronger nonlinear interactions, i.e., larger β , correspond to larger p_0 , and consequently smaller latent heat. Thus, it is possible that the first-order DPT could become a continuous phase transition in the strong coupling limit.

The critical behavior is also reflected in other physical quantities. For instance, consider the local clustering coefficient $C_i := [\sum_{j,k} A_{ij}A_{jk}A_{ki}/k_i(k_i-1)]$ for a vertex i [43,44]. Near t_c , the nonlinear interaction becomes dominant. The quartic interaction implies that the number of new triangles increases approximately linearly with the number of new edges. Consequently, the number of triangles attached to the vertex i is proportional to its degree, i.e., $\sum_{j,k} A_{ij}A_{jk}A_{ki} \sim k_i$. Equivalently, this leads to $C \sim (1/k)$ as $t \rightarrow t_c$. This theoretical prediction is confirmed by numerical simulations, as shown in Fig. 4(d). Such scaling is commonly observed in networks with modular structures, as seen in many real-world networks typically generated by hierarchical mechanisms [45,46]. Remarkably, without relying on any of these assumptions,

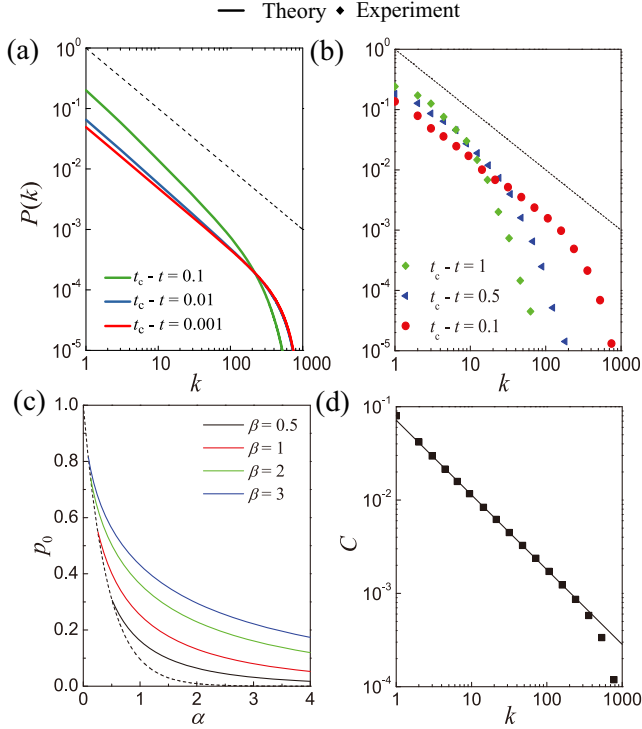


FIG. 4. Critical behavior. (a) Theoretical prediction and (b) numerical simulation of the degree distribution $P(k, t)$ for t near the critical time t_c , demonstrating that $P(k)$ follows an asymptotic power law, $P(k) \propto (1/k)$, as t approaches criticality. The dashed lines serve as a guide to the eye for the $1/k$ trend. (c) Theoretical prediction of the fraction of isolated vertices p_0 at criticality for various parameters. The dashed line indicates the phase boundary between the equilibrium and nonequilibrium phases. (d) The clustering coefficient C as a function of degree k for t near the critical time. Scatter points represent numerical results, while the solid line indicates the theoretical expectation $C(k) \propto (1/k)$.

the critical behavior of the DPT naturally results in the same law.

Discussion—We have discovered a DPT in nonequilibrium network dynamics, a phenomenon absent in well-known growing models such as triadic closure [47] and duplication models [48], which show no finite-time singularity. While our primary focus is on the nonequilibrium dynamics, the asymptotic steady states reached as $t \rightarrow \infty$ are closely related to familiar equilibrium ensembles. In the simplest linear case ($\beta = 0$) the stationary distribution coincides with the Erdős-Rényi ensemble, while with nonlinear interactions ($\beta > 0$) it connects to Strauss-type models with triangle terms [49–51] (see Supplemental Material [38], Sec. 1.3). The sparse to dense transition of these asymptotic states is consistent with Strauss condensation.

On the other hand, Strauss-type models typically collapse into uniformly dense structures, with heterogeneity visible only at criticality. In our model, by contrast,

heterogeneity arises dynamically before equilibrium is reached, leading to a finite-time singularity at a critical time t_c , fundamentally distinct from transitions previously studied in equilibrium or growth models [47–53].

This perspective also clarifies the relevance of our results for real systems. In contexts such as financial crises, epidemic outbreaks, infrastructure failures, or social unrest, interventions typically occur once instability emerges, so the system may never reach its asymptotic equilibrium. Empirical work [19] has shown that densitylike measures in online social dynamics exhibit a clear hyperbolic divergence consistent with Eq. (1). This observation is naturally explained by our theoretical prediction in Eq. (5), highlighting the central role of transient nonequilibrium dynamics and the possibility of a finite-time singularity. More broadly, our results demonstrate that dynamical network models provide a natural framework to capture finite-time critical phenomena in complex systems. Such behavior, which remains invisible from the perspective of equilibrium ensembles, may play a central role in understanding sudden instabilities across diverse real-world domains.

Finally, the emergence of DPTs in this minimal classical setting also suggests intriguing parallels with quantum dynamical phase transitions. In quantum systems, the evolution operator $U = e^{-\int H dt}$ is central to understanding phase transitions, linking nonanalyticities in the time-dependent rate function to the eigenvalues of U . In classical systems, despite the absence of unitarity, the spectrum of the evolution operator may play an analogous role. Our minimal model provides a concrete example of a classical system where DPTs emerge, shedding light on parallels with quantum DPTs. The SD equation, a powerful tool for capturing out-of-equilibrium dynamics, may act as a unifying framework for analyzing both classical and quantum DPTs, as illustrated in this Letter and in studies of quantum DPTs [54].

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J. L. and N. A. contributed equally to this work.

Data availability—The simulation code that supports the findings of this article is openly available [55].

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